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GI—22—2025

FACULTY OF SCIENCE AND TECHNOLOGY

B.C.A. (Third Year) (Fifth Semester) EXAMINATION

NOVEMBER/DECEMBER, 2025

(Revised/CBCS Pattern)

COMPUTER APPLICATION

Paper BCA-504-A

(Computer Network)

(Tuesday, 18-11-2025) **Time : 10.00 a.m. to 1.00 p.m.**

Time—3 *Hours* *Maximum Marks*—75

N.B. :— (1) *All* questions are compulsory.

(2) Figures to the right indicate full marks.

(3) Assume suitable data, if necessary.

1. Attempt any *five* of the following (**3** marks each) : 15

(*a*) Explain Ring Topology.

(*b*) Explain Bluetooth.

(*c*) Explain Microwave.

P.T.O.

(*d*) Difference between LAN and WAN.

(*e*) What is Computer Network ?

(*f*) Write a short note on DNS.

(*g*) Write a short note on URL.

2. Attempt any *three* of the following (**5** marks each) : 15

(*a*) Explain IP address.

(*b*) Explain Traditional Cryptography in detail.

(*c*) Explain OSI reference model.

(*d*) Explain Twisted pair cable in detail.

(*e*) Explain Transposition Ciphers in detail.

3. Attempt any *three* of the following (**5** marks each) : 15

(*a*) Explain Data Encryption Standard (DES).

(*b*) Explain TCP/IP reference model.

(*c*) Explain Network Security issues in detail.

(*d*) Explain Fiber optic cable in detail.

(*e*) Explain Intranet and Extranet in detail.

4. Attempt any *three* of the following (**5** marks each) : 15

(*a*) Explain DHCP in detail.

(b) What is Ethernet ? Explain types of Ethernet.

(c) Explain Internet Architected in detail.

(d) What is Topology ? Explain Bus and Mesh topology.

(e) Explain FTP & HTTP in detail.

5. Write short notes on any *three* of the following (**5** marks each) : 15

(a) Radio Transmission

(b) Wireless LAN

(c) Firewall

(d) Digital signature

(e) ISP.

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